Name: D S LAKSHMI NARAYANA

Register No.: 192512371

Course Name: SIGNALS AND SYSTEM

Course Code: ECA 0304

Experiment 15: Discrete-Time Fourier Transform and Frequency-Domain Analysis

OBJECTIVES

• Compute the DTFT of finite-length discrete-time sequences.

• Plot and interpret DTFT magnitude and phase spectra.

• Verify DTFT periodicity with period 2π and key properties including linearity and time shift.

• Relate sequence length to main-lobe width and spectral spreading.

Experimental Procedure

1. Define a length-10 rectangular pulse $x[n]=1$ for $0 \le n \le N-1$.
2. Evaluate the DTFT by direct summation on a dense frequency grid.
3. Plot magnitude and phase spectra over one period.
4. Plot magnitude over two periods to verify 2π periodicity.
5. Shift the sequence and compare magnitude and phase.
6. Combine two sequences to verify linearity.
7. Vary N and observe the main-lobe width.

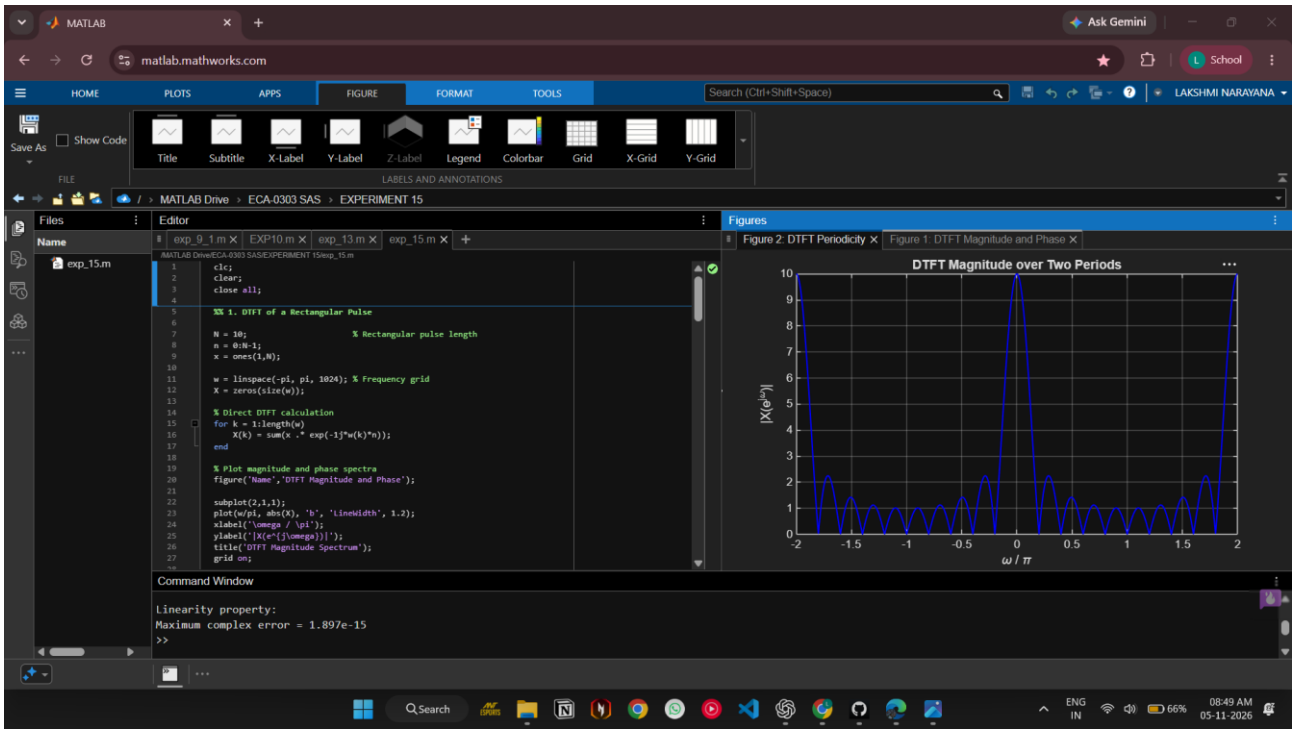
MATLAB Coding

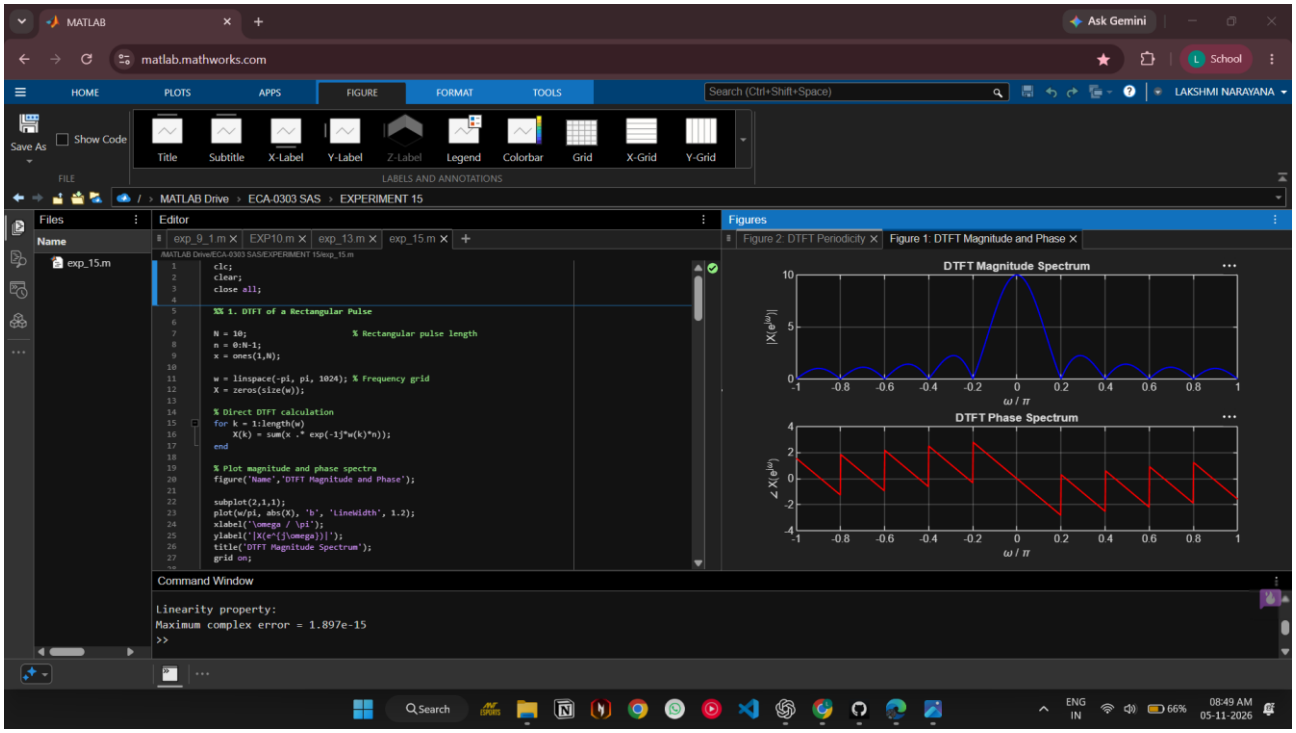
```
clc; clear; close all;
N = 10;                 % rectangular pulse length
n = 0:N-1;
x = ones(1,N);
w = linspace(-pi, pi, 1024);   % dense frequency grid (one period)
X = zeros(size(w));
for k = 1:length(w)
X(k) = sum(x .* exp(-1j*w(k)*n));   % direct DTFT summation
end
```

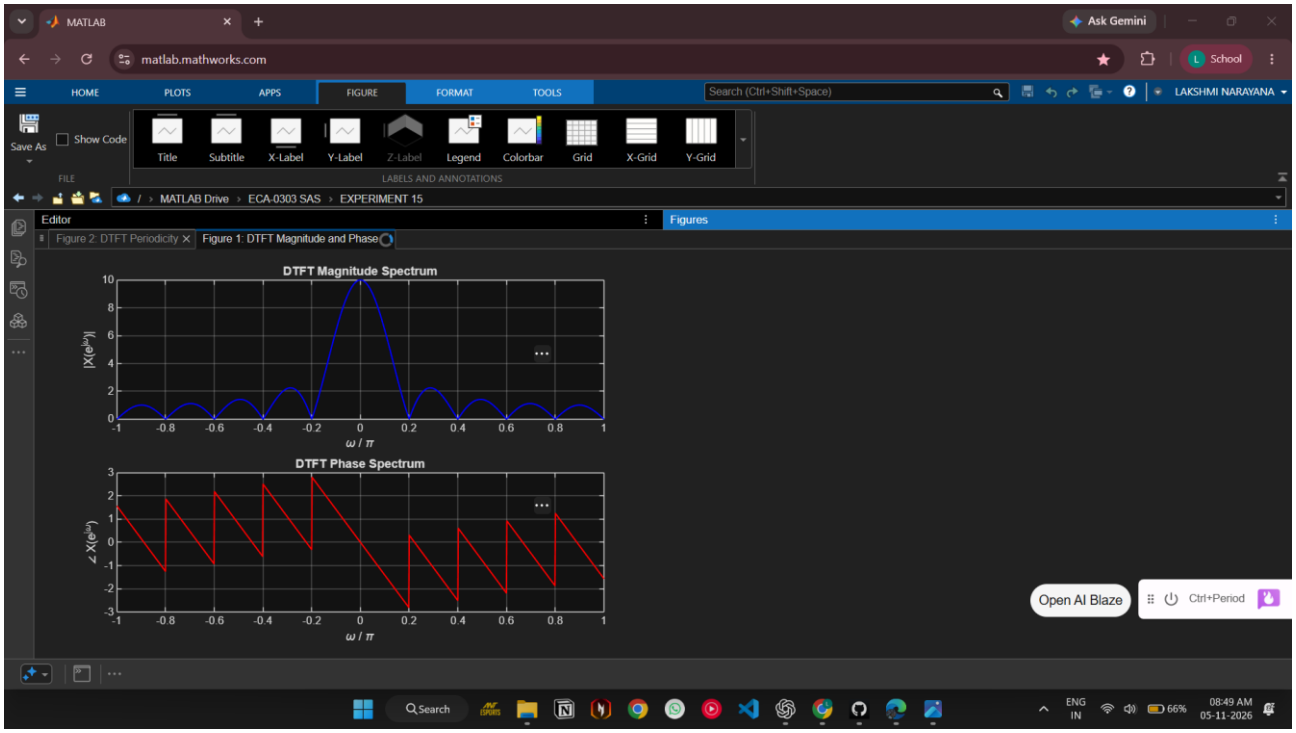
```
figure;
subplot(2,1,1); plot(w/pi, abs(X)); xlabel('\omega / \pi'); ylabel('|X(e^{j\omega})|');
title('DTFT Magnitude Spectrum'); grid on;
subplot(2,1,2); plot(w/pi, angle(X)); xlabel('\omega / \pi'); ylabel('\angle X(e^{j\omega})');
title('DTFT Phase Spectrum'); grid on;
%% Periodicity check over 2 periods
w2 = linspace(-2*pi, 2*pi, 2048);
X2 = zeros(size(w2));
for k = 1:length(w2), X2(k) = sum(x .* exp(-1j*w2(k)*n)); end
figure; plot(w2/pi, abs(X2)); xlabel('\omega / \pi'); ylabel('|X(e^{j\omega})|');
title('DTFT Magnitude over Two Periods (Periodicity = 2\pi)'); grid on;
%% Time-shift property verification
n0 = 3;
x_shift = [zeros(1,n0), x];
n_shift = 0:length(x_shift)-1;
Xs = zeros(size(w));
for k = 1:length(w), Xs(k) = sum(x_shift .* exp(-1j*w(k)*n_shift)); end
mag_err = max(abs(abs(Xs) - abs(X)));
phase_theory = -w*n0;
phase_err = max(abs(mod(angle(Xs)-angle(X)-phase_theory+pi,2*pi)-pi));
fprintf('Time-shift property: max magnitude error = %.4g\n', mag_err);
fprintf('Time-shift property: max phase-term error (rad) = %.4g\n', phase_err);
%% Linearity property verification
a = 2; b = -1;
x1 = ones(1,5); x2 = [1 0.8 0.6 0.4 0.2 0 0 0];
n1 = 0:length(x1)-1; n2 = 0:length(x2)-1;
xc = a*[x1 zeros(1,length(x2)-length(x1))] + b*x2;
nc = 0:length(xc)-1;
X1 = zeros(size(w)); X2v = zeros(size(w)); Xcomb = zeros(size(w));
for k = 1:length(w)
X1(k)    = sum(x1 .* exp(-1j*w(k)*n1));
X2v(k)   = sum(x2 .* exp(-1j*w(k)*n2));
Xcomb(k) = sum(xc .* exp(-1j*w(k)*nc));
```

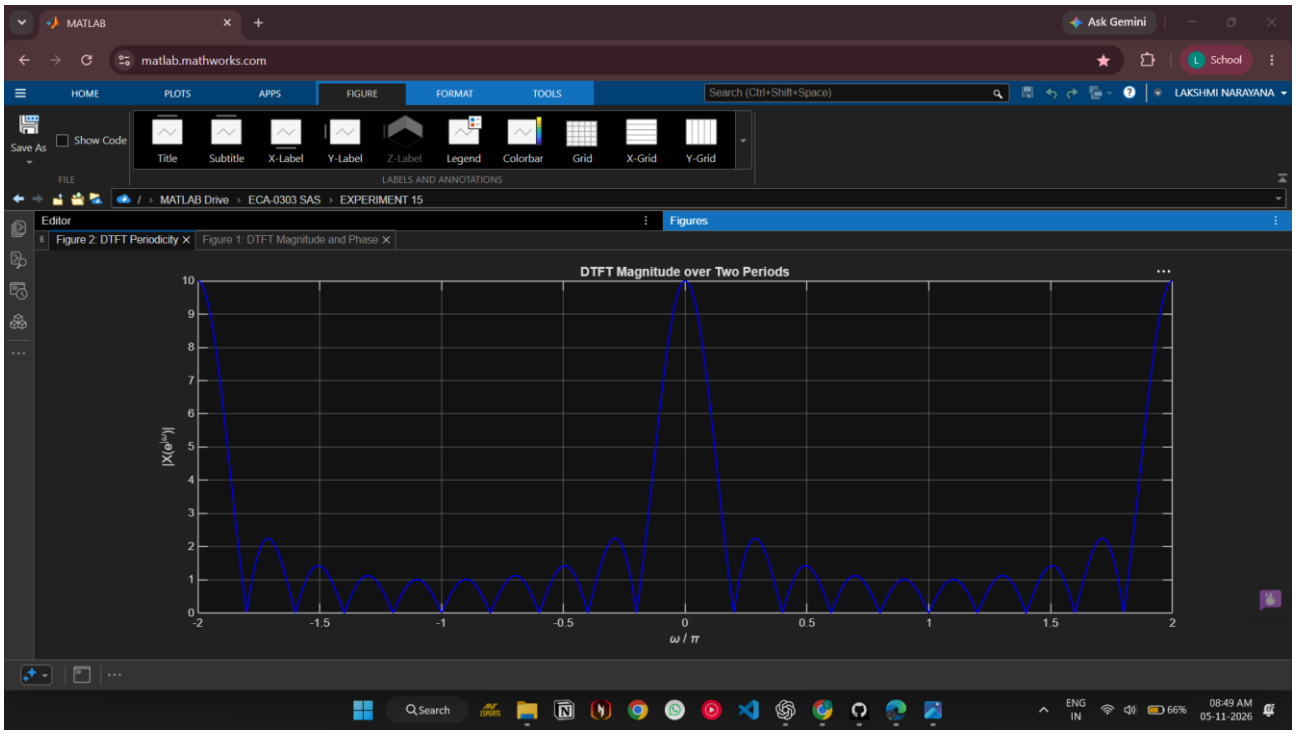
```
end
lin_err = max(abs(Xcomb - (a*X1 + b*X2v)));
fprintf('Linearity property: max complex error = %.4g\n', lin_err);
```

Input and Output Graphs






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Figures
Figure 2: DTFT Periodicity
Figure 1: DTFT Magnitude and Phase
DTFT Magnitude Spectrum
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Editor
Figures
Figure 2: DTFT Periodicity
Figure 1: DTFT Magnitude and Phase
DTFT Magnitude over Two Periods
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